
Arakelov Geometry

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1 Classical height machine

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3 Hermitian line bundles

Let K be a number field with ring of integers $\mathcal{O}_K$. Let $\mathcal{X}$ be an arithmetic variety over $\text{Spec } \mathcal{O}_K$ of relative dimension n . Yang: flat, separated, of finite type,

3.1 Definitions and examples

Definition 3.1. A hermitian line bundle $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$ on $\mathcal{X}$ consists of

- a line bundle $\mathcal{L}$ on $\mathcal{X}$;
- for each embedding $\sigma : K \hookrightarrow \mathbb{C}$ (equivalently, each infinite place $\sigma \in M_K^\infty$), a smooth hermitian metric $\|\cdot\|_\sigma$ on the complex line bundle $\mathcal{L}_\sigma(\mathbb{C})$ over $\mathcal{X}_\sigma(\mathbb{C})$, which is invariant under the complex conjugation map.

Yang: To be checked.

Example 3.2. Let $\mathcal{X} = \mathbb{P}_\mathbb{Z}^n$.

Yang: To be continued.

Definition 3.3. Let $\overline{\mathcal{L}}_1$ and $\overline{\mathcal{L}}_2$ be hermitian line bundles on $\mathcal{X}$.

- The tensor product $\overline{\mathcal{L}}_1 \otimes \overline{\mathcal{L}}_2$ is defined as the line bundle $\mathcal{L}_1 \otimes \mathcal{L}_2$ on $\mathcal{X}$ equipped with the metrics $\|s_1 \otimes s_2\|_\sigma = \|s_1\|_{\sigma,1} \cdot \|s_2\|_{\sigma,2}$ on $(\mathcal{L}_1 \otimes \mathcal{L}_2)_\sigma(\mathbb{C})$
- The dual $\overline{\mathcal{L}}_1^\vee$ is defined as the dual line bundle $\mathcal{L}_1^\vee$ on $\mathcal{X}$ equipped with the dual metrics $\|\cdot\|_{\sigma,1}^\vee$

on $(\mathcal{L}_1^\vee)_\sigma(\mathbb{C})$ for each $\sigma \in M_K^\infty$.

Yang: To be checked.

Definition 3.4. Let $f : \mathcal{Y} \rightarrow \mathcal{X}$ be a morphism of arithmetic varieties over $\text{Spec } \mathcal{O}_K$. Let $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$ be a hermitian line bundle on $\mathcal{X}$. The *pullback* $f^*\overline{\mathcal{L}}$ is defined as the line bundle $f^*\mathcal{L}$ on $\mathcal{Y}$ equipped with the metrics $\|s\|'_\sigma = \|f_\sigma^*s\|_\sigma$ on $(f^*\mathcal{L})_\sigma(\mathbb{C})$ for each $\sigma \in M_K^\infty$, where $f_\sigma : \mathcal{Y}_\sigma(\mathbb{C}) \rightarrow \mathcal{X}_\sigma(\mathbb{C})$ is the induced morphism of complex manifolds. Yang: To be checked.

Definition 3.5. Let $\overline{\mathcal{L}} = (\mathcal{L}, \{\|\cdot\|_\sigma\}_{\sigma \in M_K^\infty})$ be a hermitian line bundle on $\mathcal{X}$. The set of *small sections* of $\overline{\mathcal{L}}$ is defined as

$$H^0(\mathcal{X}, \overline{\mathcal{L}}) := \{s \in H^0(\mathcal{X}, \mathcal{L}) : \|s\|_\sigma \leq 1 \text{ for all } \sigma \in M_K^\infty\}.$$

Yang: To be checked.

Proposition 3.6. The set $H^0(\mathcal{X}, \overline{\mathcal{L}})$ is finite.

Definition 3.7. Let $\overline{\mathcal{L}}$ be a hermitian line bundle on $\mathcal{X}$. We set

$$h^0(\mathcal{X}, \overline{\mathcal{L}}) := \log \#H^0(\mathcal{X}, \overline{\mathcal{L}}).$$

The *volume* of $\overline{\mathcal{L}}$ is defined as

$$\text{vol}(\overline{\mathcal{L}}) := \limsup_{m \rightarrow \infty} \frac{h^0(\mathcal{X}, \overline{\mathcal{L}}^{\otimes m})}{m^{n+1}/(n+1)!}.$$

4 Arithmetic Hilbert-Samuel theorem

Theorem 4.1. Let $\mathcal{X}$ be an arithmetic variety of absolute dimension d over $\text{Spec } \mathcal{O}_K$, and let $\overline{\mathcal{L}}$ be a hermitian line bundle on $\mathcal{X}$. Assume that $\overline{\mathcal{L}}$ is nef. Then, we have

$$\text{vol}(\overline{\mathcal{L}}) = \widehat{\deg}(\hat{c}_1(\overline{\mathcal{L}})^d).$$

5 Arithmetic fundamental inequality